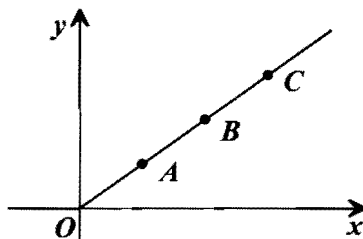




Year 11 Mathematics Extension Task 2
Friday 24th June 2011
Time Allowed: 70 minutes

- Instructions:
- a) Write all your answers on the paper provided.
 - b) Show all necessary working.
 - c) Marks may be deducted for careless or badly arranged work.
 - d) Use black or blue pen to write your solutions.
 - e) No liquid paper is to be used. If a correction is to be made, rule one line through the incorrect solution.

1. In the diagram the points $A(1,1)$, $B(2,2)$, $C(3,3)$ and the Origin are illustrated.



2

Indicate whether each of the following is Correct or Incorrect.

- a. A divides the interval OC internally in the ratio $1:2$
- b. A divides the interval BC externally in the ratio $1:2$
- c. B divides the interval CO internally in the ratio $1:2$
- d. C divides the interval AB externally in the ratio $1:2$

2. Find the acute angle between the lines $x - y + 3 = 0$ and $2x + y + 1 = 0$. Give your answer correct to the nearest minute.

3

3. Without using your calculator, and showing all of your working, find the exact values of;

- a) $\sin 105^\circ$
- b) $\frac{1 - \tan^2 15^\circ}{1 + \tan^2 15^\circ}$

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4. If X and Y are acute angles such that $\tan X = \frac{5}{12}$ and $\cos Y = \frac{3}{\sqrt{13}}$, find the exact values of;

- a) $\cos(X - Y)$
- b) $\tan \frac{X}{2}$

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5. Let $A(-1, 2)$ and $B(3, 5)$ be points on the number plane. Find the coordinates of the point C that divides the interval AB internally in the ratio $1:5$

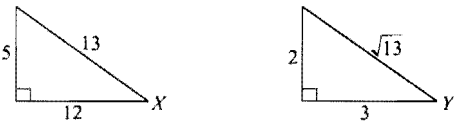
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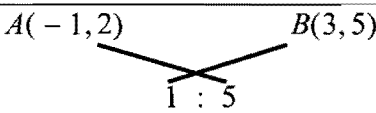
6. The lines $3x - y + 2 = 0$ and $mx - y - 1 = 0$ intersect at 60° . Find the possible values of m .

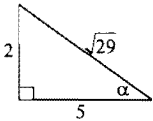
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7.	Prove the following; a) $\frac{\cos 2x}{\cos x + \sin x} = \cos x - \sin x$ b) $2 - 2\tan x - \frac{2\tan x}{\tan 2x} = (1 - \tan x)^2$ c) $\frac{\tan A - \tan B}{\tan A + \tan B} = \frac{\sin(A - B)}{\sin(A + B)}$	2 2 3
8.	If $t = \tan \frac{x}{2}$, express $\sqrt{\frac{1 - \sin x}{1 + \sin x}}$ in terms of t .	3
9.	The point $P(-12, -2)$ divides the interval joining $A(-2, 4)$ and $B(3, 7)$, externally in the ratio $k : 1$. Find the value of k .	2
10.	Solve the following, correct to the nearest degree, for $0^\circ \leq x \leq 360^\circ$ a) $2\cos x + 5\sin x = 3$, by first rewriting as a single trigonometric function. b) $3\cos x + 4\sin x = 4$, using the t -results.	3 3
11.	Explain, in your own words, why it is important to check whether $x = 180^\circ$ is a solution when using the t -results to solve a trigonometric equation.	1
12.	a) Express the equation $\operatorname{cosec} \theta (3\cos 2\theta + 7) + 11 = 0$, in the form $a\sin^2 \theta + b\sin \theta + c = 0$, where a , b and c are constants. b) Hence solve for $0^\circ \leq \theta \leq 360^\circ$, the equation $\operatorname{cosec} \theta (3\cos 2\theta + 7) + 11 = 0$, correct to the nearest degree.	2 2
13.	a) Show that $\tan r x \tan(r + 1)x = \frac{\tan(r + 1)x}{\tan x} - \frac{\tan r x}{\tan x} - 1$ b) Hence, or otherwise, show that $\tan 20^\circ \tan 40^\circ + \tan 40^\circ \tan 60^\circ + \dots + \tan 180^\circ \tan 200^\circ = -9$	2 2
END OF EXAM		

BAULKHAM HILLS HIGH SCHOOL
YEAR 11 EXTENSION TASK 2 2011 SOLUTIONS

Solution	Marks	Comments
1. a) Correct b) Correct c) Correct d) Incorrect	2	1 mark • 3 out of 4 correct
2. $x - y + 3 = 0 \Rightarrow m = 1$ $2x + y + 1 = 0 \Rightarrow m = -2$ $\tan \alpha = \left \frac{1 - (-2)}{1 + (1) \times (-2)} \right $ $= \left \frac{3}{-1} \right $ $= 3$ $\alpha = 71^{\circ}34'$	3	2 marks • $\tan \alpha = 3$ • Correct solution for their slopes 1 mark • Correctly finds the two required slopes • Correctly substitutes their slopes into the formula <i>Note: no penalty for rounding error</i>
3a. $\sin 105^{\circ} = \sin(60 + 45)^{\circ}$ $= \sin 60^{\circ} \cos 45^{\circ} + \cos 60^{\circ} \sin 45^{\circ}$ $= \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{\sqrt{2}}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{\sqrt{2}}\right)$ $= \frac{\sqrt{3} + 1}{2\sqrt{2}}$	2	1 mark • Correctly applies $\sin(\alpha + \beta)$ disregarding actual values substituted
3b. $\frac{1 - \tan^2 15^{\circ}}{1 + \tan^2 15^{\circ}} = \cos(2 \times 15)^{\circ}$ $= \cos 30^{\circ}$ $= \frac{\sqrt{3}}{2}$	2	1 mark • Recognises t result for $\cos \theta$
4a.  $\cos(X - Y) = \cos X \cos Y + \sin X \sin Y$ $= \left(\frac{12}{13}\right)\left(\frac{3}{\sqrt{13}}\right) + \left(\frac{5}{13}\right)\left(\frac{2}{\sqrt{13}}\right)$ $= \frac{46}{13\sqrt{13}}$	2	1 mark • Correctly applies $\cos(\alpha - \beta)$

Solution		Marks	Comments
4b.	$\tan X = \frac{2 \tan \frac{X}{2}}{1 - \tan^2 \frac{X}{2}}$ $\frac{5}{12} = \frac{2 \tan \frac{X}{2}}{1 - \tan^2 \frac{X}{2}}$ $5 - 5 \tan^2 \frac{X}{2} = 24 \tan \frac{X}{2}$ $5 \tan^2 \frac{X}{2} + 24 \tan \frac{X}{2} - 5 = 0$ $\left(5 \tan \frac{X}{2} - 1 \right) \left(\tan \frac{X}{2} + 5 \right) = 0$ $\tan \frac{X}{2} = \frac{1}{5} \quad \text{or} \quad \tan \frac{X}{2} = -5$ However $\frac{X}{2}$ is acute $\therefore \tan \frac{X}{2} = \frac{1}{5}$	3	2 marks • Correct solution, however, does not justify why negative solution is discounted • Solves the quadratic correctly • Solves a quadratic, gaining two solutions and justifies the discounting of one solution. 1 mark • Uses a correct method
5.	 $C = \left(\frac{-1 \times 5 + 3 \times 1}{1 + 5}, \frac{2 \times 5 + 5 \times 1}{1 + 5} \right)$ $= \left(-\frac{1}{3}, \frac{5}{2} \right)$	2	1 mark • Uses a correct method
6.	$3x - y + 2 = 0 \Rightarrow m = 3$ $\tan 60^\circ = \left \frac{m - 3}{1 + 3m} \right $ $\sqrt{3} = \left \frac{m - 3}{1 + 3m} \right $ $\sqrt{3} 1 + 3m = m - 3 $ $\sqrt{3} (1 + 3m) = m - 3 \qquad -\sqrt{3} (1 + 3m) = m - 3$ $(1 - 3\sqrt{3})m = 3 + \sqrt{3} \qquad (1 + 3\sqrt{3})m = 3 - \sqrt{3}$ $m = \frac{3 + \sqrt{3}}{1 - 3\sqrt{3}} \qquad \text{or} \qquad m = \frac{3 - \sqrt{3}}{1 + 3\sqrt{3}}$	3	2 marks • Correctly finds one possible value of m • Finds two values for m, correct for their value of $\tan 60^\circ$. • Establishes a correct absolute value equation 1 mark • Uses a correct method
7a.	$\frac{\cos 2x}{\cos x + \sin x} = \frac{\cos^2 x - \sin^2 x}{\cos x + \sin x}$ $= \frac{(\cos x + \sin x)(\cos x - \sin x)}{(\cos x + \sin x)}$ $= \cos x - \sin x$	2	1 mark • Progress towards solution involving correct substitution of $\cos 2x$ or equivalent merit
7b.	$2 - 2 \tan x - \frac{2 \tan x}{\tan 2x} = 2 - 2 \tan x - \frac{2 \tan x}{1} \times \frac{1 - \tan^2 x}{2 \tan x}$ $= 2 - 2 \tan x - 1 + \tan^2 x$ $= 1 - 2 \tan x + \tan^2 x$ $= (1 - \tan x)^2$	2	1 mark • Progress towards solution involving correct substitution of $\tan 2x$ or equivalent merit

Solution		Marks	Comments
7c.	$\frac{\tan A - \tan B}{\tan A + \tan B} = \frac{\frac{\sin A}{\cos A} - \frac{\sin B}{\cos B}}{\frac{\sin A}{\cos A} + \frac{\sin B}{\cos B}}$ $= \frac{\frac{\sin A \cos B - \cos A \sin B}{\cos A \cos B}}{\frac{\sin A \cos B + \cos A \sin B}{\cos A \cos B}}$ $= \frac{\sin(A - B)}{\sin(A + B)}$	3	2 marks - Significant progress towards correct solution 1 mark - Progress towards solution that involves either correct substitution for tan's or sin (A+B) or equivalent merit <i>Note: in parts a to c, any attempt to solve like an equation should only be marked up to that point.</i>
8.	$\sqrt{\frac{1 - \sin x}{1 + \sin x}} = \sqrt{\frac{1 - \frac{2t}{1+t^2}}{1 + \frac{2t}{1+t^2}}}$ $= \sqrt{\frac{1+t^2-2t}{1+t^2+2t}}$ $= \sqrt{\frac{(1-t)^2}{(1+t)^2}}$ $= \left \frac{1-t}{1+t} \right $	3	2 marks - Significant progress towards correct solution 1 mark - Progress towards solution that involves correct t result for sinx <i>Note: no penalty incurred for the lack of absolute value signs.</i>
9.	$A(-2, 4) \quad B(3, 7)$ $-k : 1$ Using x-values: $\frac{-2 \times 1 + 3 \times -k}{-k + 1} = -12$ $-2 - 3k = 12k - 12$ $15k = 10$ $k = \frac{2}{3}$	2	1 mark - Uses a correct method $k = -\frac{2}{3}$
10a.	 $\alpha = \tan^{-1}\left(\frac{2}{5}\right)$ $= 21.8^\circ$ $2\cos x + 5\sin x = 3$ $\sqrt{29} \sin(x + 21.8)^\circ = 3$ $\sin(x + 21.8)^\circ = \frac{3}{\sqrt{29}}$ Q1 & Q2 $x + 21.8^\circ = 33.9^\circ, 146.1^\circ$ $\therefore x = 12^\circ, 124^\circ$	3	2 marks - Correctly finds two possible answers for the "compound" angle - Correctly finds two possible values of x, considering their answers for the compound angle 1 mark - Correctly transforms into a single trig expression

	Solution	Marks	Comments
10.	$3\cos x + 4\sin x = 4$ $3\left(\frac{1-t^2}{1+t^2}\right) + 4\left(\frac{2t}{1+t^2}\right) = 4$ $3 - 3t^2 + 8t = 4 + 4t^2$ $7t^2 - 8t + 1 = 0$ $(7t-1)(t-1) = 0$ $t = \frac{1}{7} \quad t = 1$ $\tan\frac{x}{2} = \frac{1}{7} \quad \tan\frac{x}{2} = 1$ $\text{Q1} \quad \text{Q1}$ $\frac{x}{2} = 8.1^\circ \quad \frac{x}{2} = 45^\circ$ $x = 16.2^\circ \quad \text{or} \quad x = 90^\circ$ $\therefore x = 16^\circ, 90^\circ$	3	2 marks • Finds more than two values for x • Correctly evaluates one possible value for x • Correctly finds two possibilities for $\tan\frac{x}{2}$ • Correctly finds two possible values for x from their quadratic 1 mark • Correctly substitutes the t-results
11.	As $\tan 90^\circ$ is undefined, which is a solution that cannot be found when solving a polynomial. i.e. the t -results method will never discover the solution $x = 180^\circ$, so its possibility must be checked.	1	
12a.	$\operatorname{cosec}\theta(3\cos 2\theta + 7) + 11 = 0$ $\frac{3(1-2\sin^2\theta) + 7}{\sin\theta} + 11 = 0$ $\frac{6\sin^2\theta - 10}{\sin\theta} - 11 = 0$ $6\sin^2\theta - 11\sin\theta - 10 = 0$	2	1 mark • Uses $\cos 2\theta = 1 - 2\sin^2\theta$
12b.	$\operatorname{cosec}\theta(3\cos 2\theta + 7) + 11 = 0$ $6\sin^2\theta - 11\sin\theta - 10 = 0$ $(3\sin\theta + 2)(2\sin\theta - 5) = 0$ $\sin\theta = -\frac{2}{3} \quad \sin\theta = \frac{5}{2}$ $\text{Q3 \& Q4} \quad \text{no solutions}$ $\theta = 222^\circ, 318^\circ \quad \text{or}$ $\therefore \theta = 222^\circ, 318^\circ$	2	1 mark • Correctly finds two possible values for $\sin x$ • Correct two values for θ, without evidence that $\sin\theta = \frac{5}{2}$ has been considered
13a.	$\tan x = \tan\{(r+1)x - rx\}$ $= \frac{\tan(r+1)x - \tan rx}{1 + \tan(r+1)x \tan rx}$ $\therefore 1 + \tan(r+1)x \tan rx = \frac{\tan(r+1)x - \tan rx}{\tan x}$ $\tan(r+1)x \tan rx = \frac{\tan(r+1)x}{\tan x} - \frac{\tan rx}{\tan x} - 1$	2	1 mark • Progress towards a solution, involving the use of $\tan(A-B)$ expansion
13b.	$\tan 20^\circ \tan 40^\circ + \tan 40^\circ \tan 60^\circ + \dots + \tan 180^\circ \tan 200^\circ$ $= \frac{\tan 40^\circ}{\tan 20^\circ} - \frac{\tan 20^\circ}{\tan 20^\circ} - 1 + \frac{\tan 60^\circ}{\tan 20^\circ} - \frac{\tan 40^\circ}{\tan 20^\circ} - 1 + \dots + \frac{\tan 200^\circ}{\tan 20^\circ} - \frac{\tan 180^\circ}{\tan 20^\circ} - 1$ $= \frac{\tan 200^\circ}{\tan 20^\circ} - \frac{\tan 20^\circ}{\tan 20^\circ} - 9 \times 1$ $= \frac{\tan 20^\circ}{\tan 20^\circ} - \frac{\tan 20^\circ}{\tan 20^\circ} - 9$ $= 1 - 1 - 9$ $= -9$	2	1 mark • Correctly simplifying the sum to three terms.